Power BI Overview

REACH Webinar Series 2026 - Session One

// Robert Pilgrim, Ph.D.

On-line version of these slides

https://rppilgrim.github.io/website-REACH-2026-Power-BI-Webinar/

Who am I

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Purpose of these Webinars

- Met many people:
  - Conferences / Summits
  - Workshops / presentations
  - REACH mentoring
- Most people just need help starting

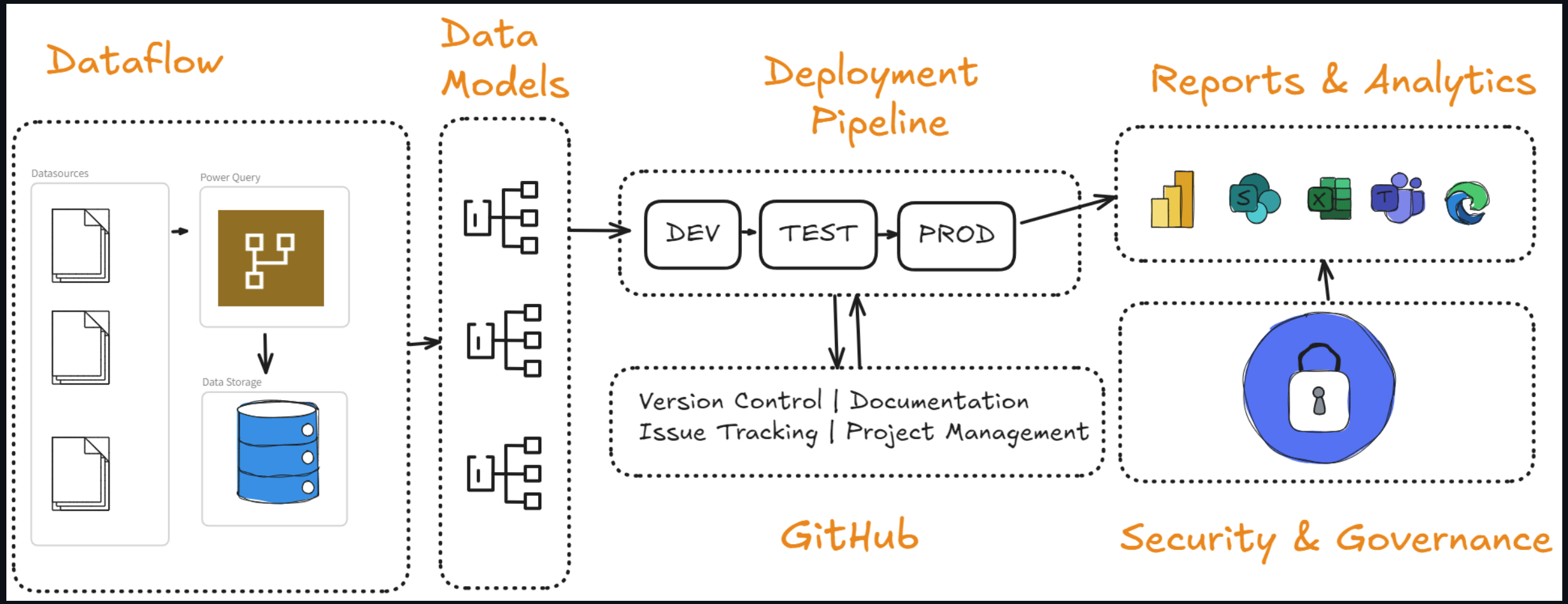
A Note on Data

- BIG friction "no data access"
- Dummy data or public data (ASU)
- No direct system access needed
- Federal: HERD, IPEDS, SBIR/STTR
- Let's get started...

Session Framework

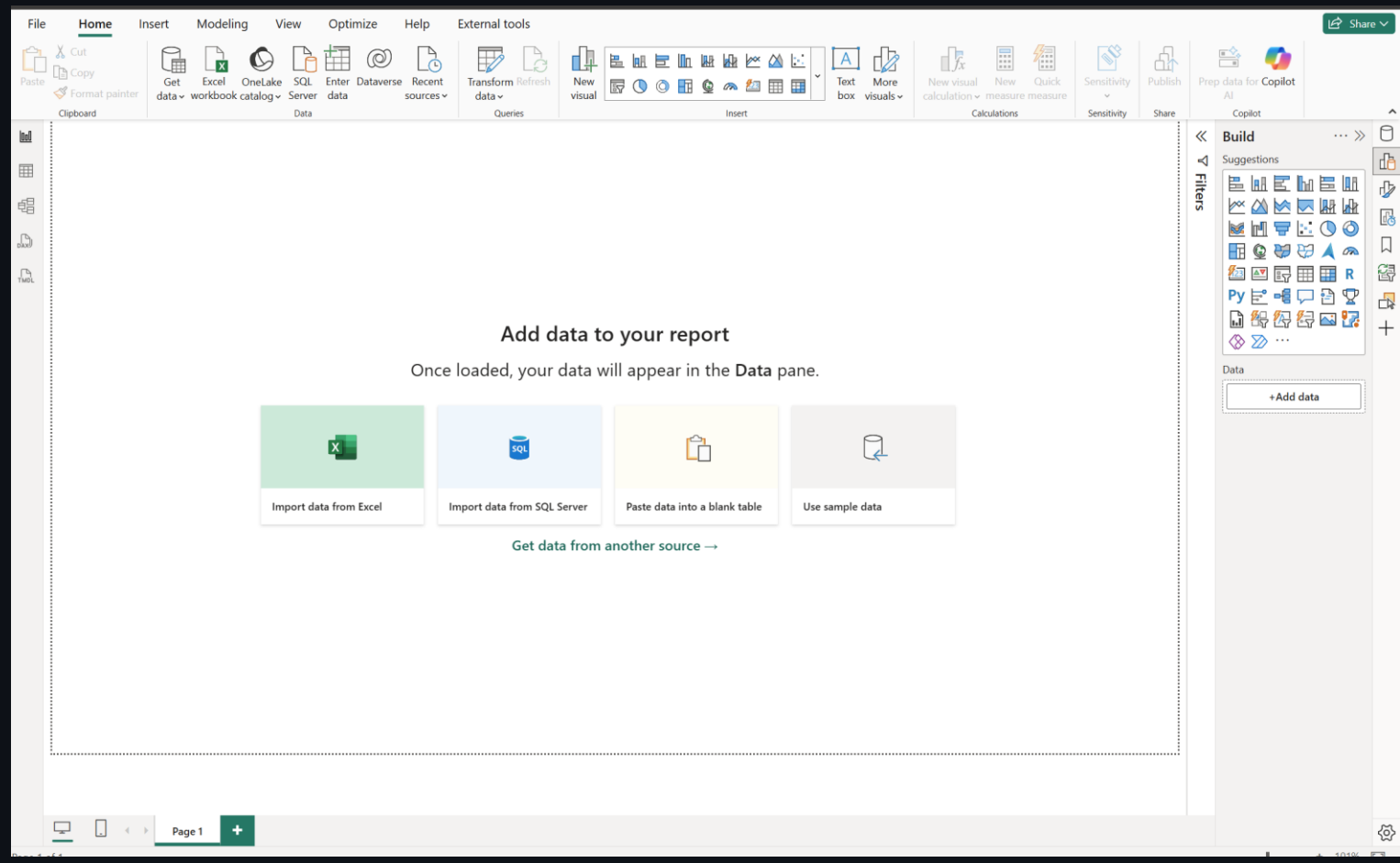
One: Power BI Overview	Two: Security and Teams	Three: Advanced Concepts
• Introduction • Microsoft Analytics Core Concepts • Start on our Power BI Report	• Continue report • Deploy Report • Sharing & Security • Version Control • Documentation	• Finish our Report • User Defined Functions & Parameters • R and Python • External Tools

Simplified Data Workflow



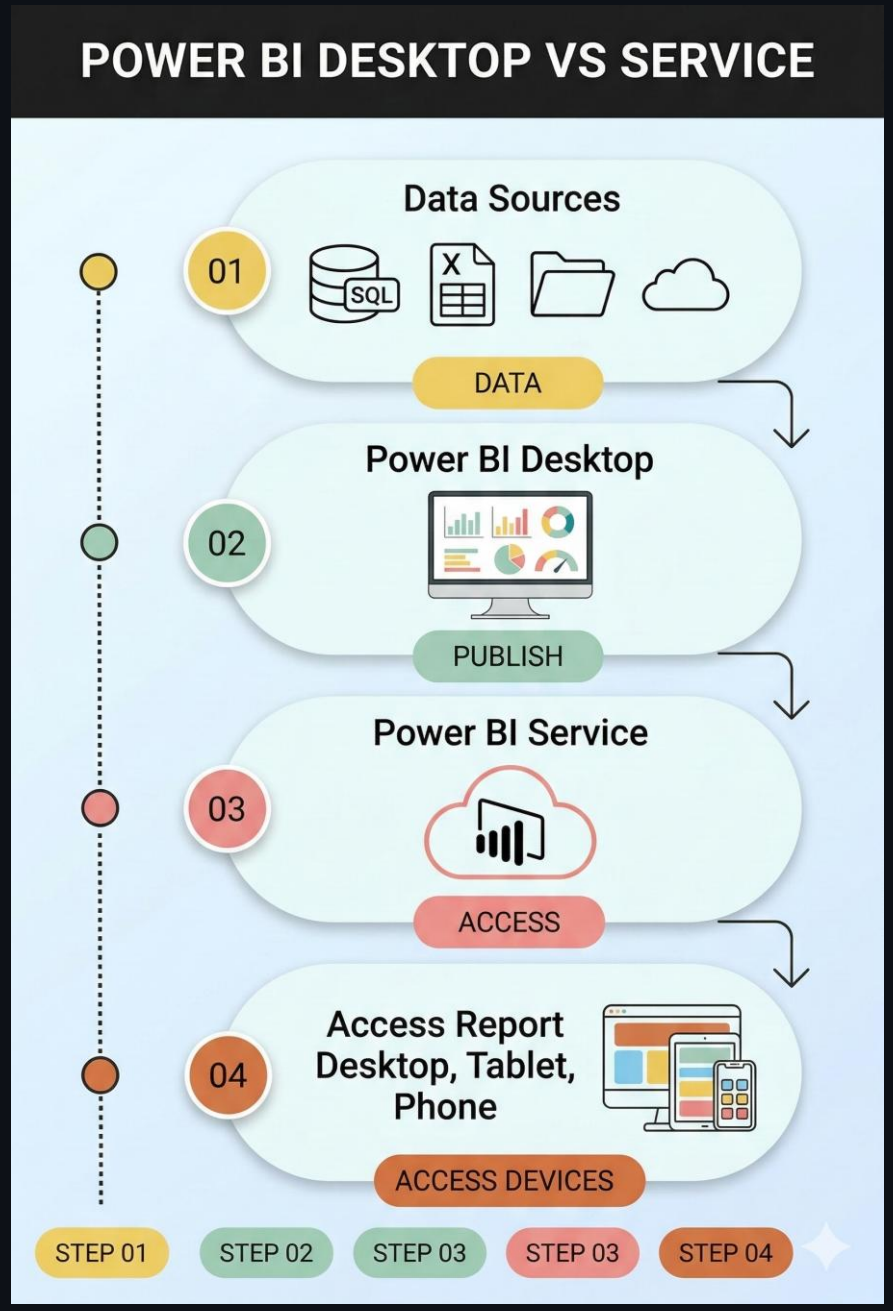
Power BI Desktop

- 100% Free, no restrictions
- Why need a license?
- Publishing, sharing, collaboration



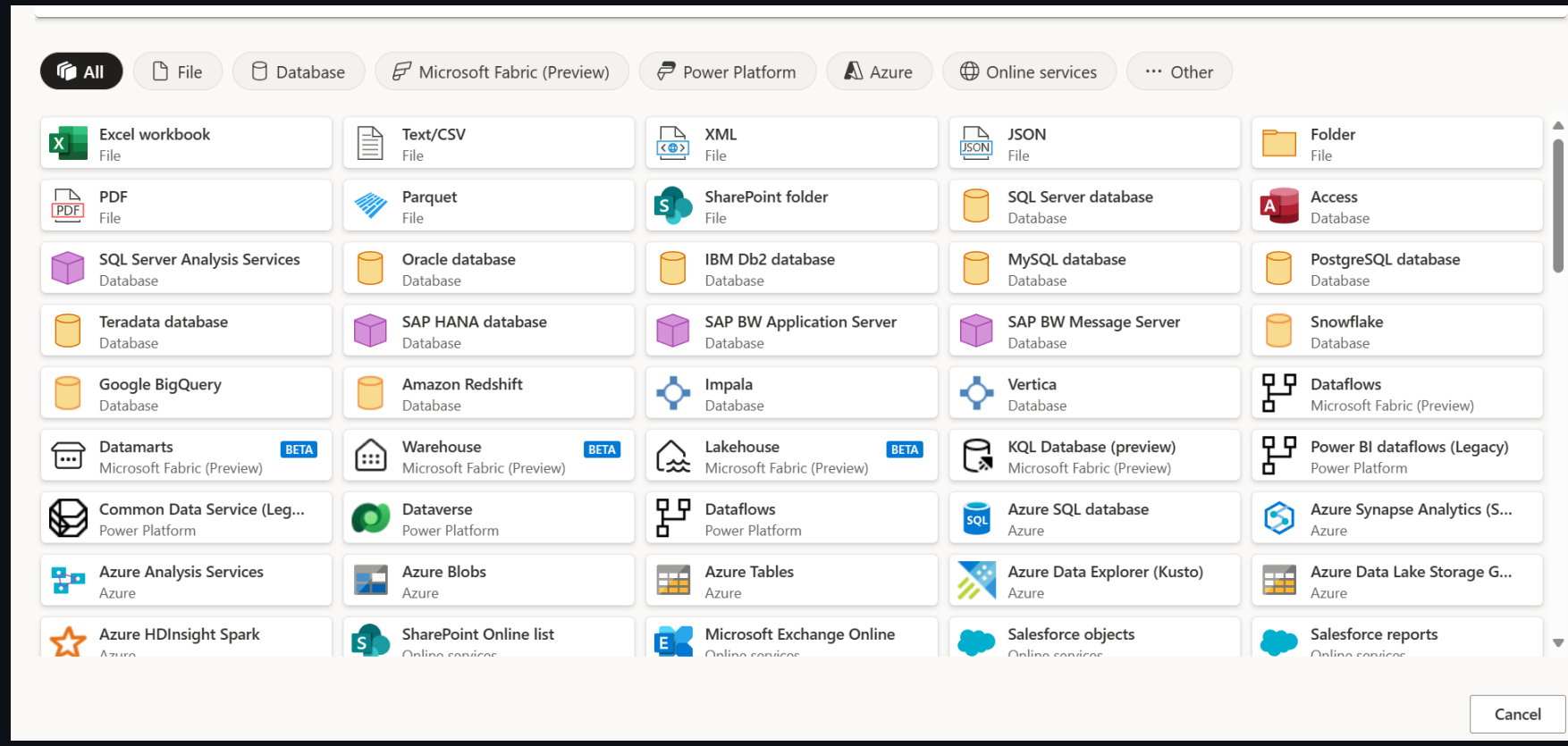
Power BI Service

- Cloud based
- Share / Collaborate
- Microsoft campus?
- app.powerbi.com



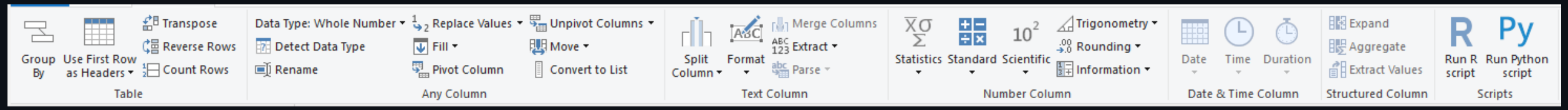
Power Query Get Data

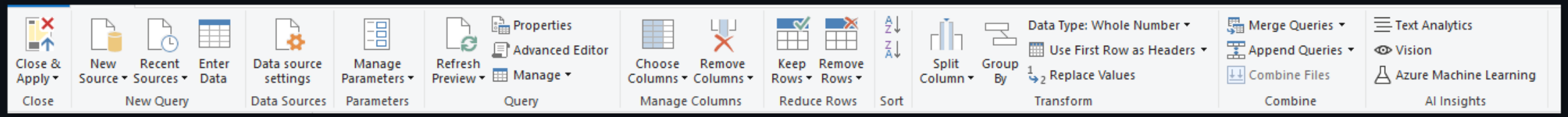
- Wide range of data sources
- Same cloud / desktop / Excel
- Code / No-code

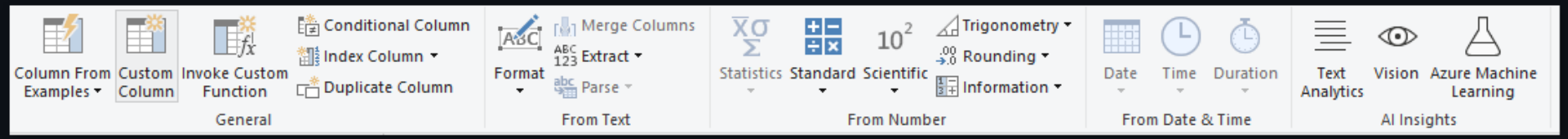


Power Query Transform your Data

- Connect , Clean, Transform and Merge
- Code directly or GUI
- Python, R and SQL

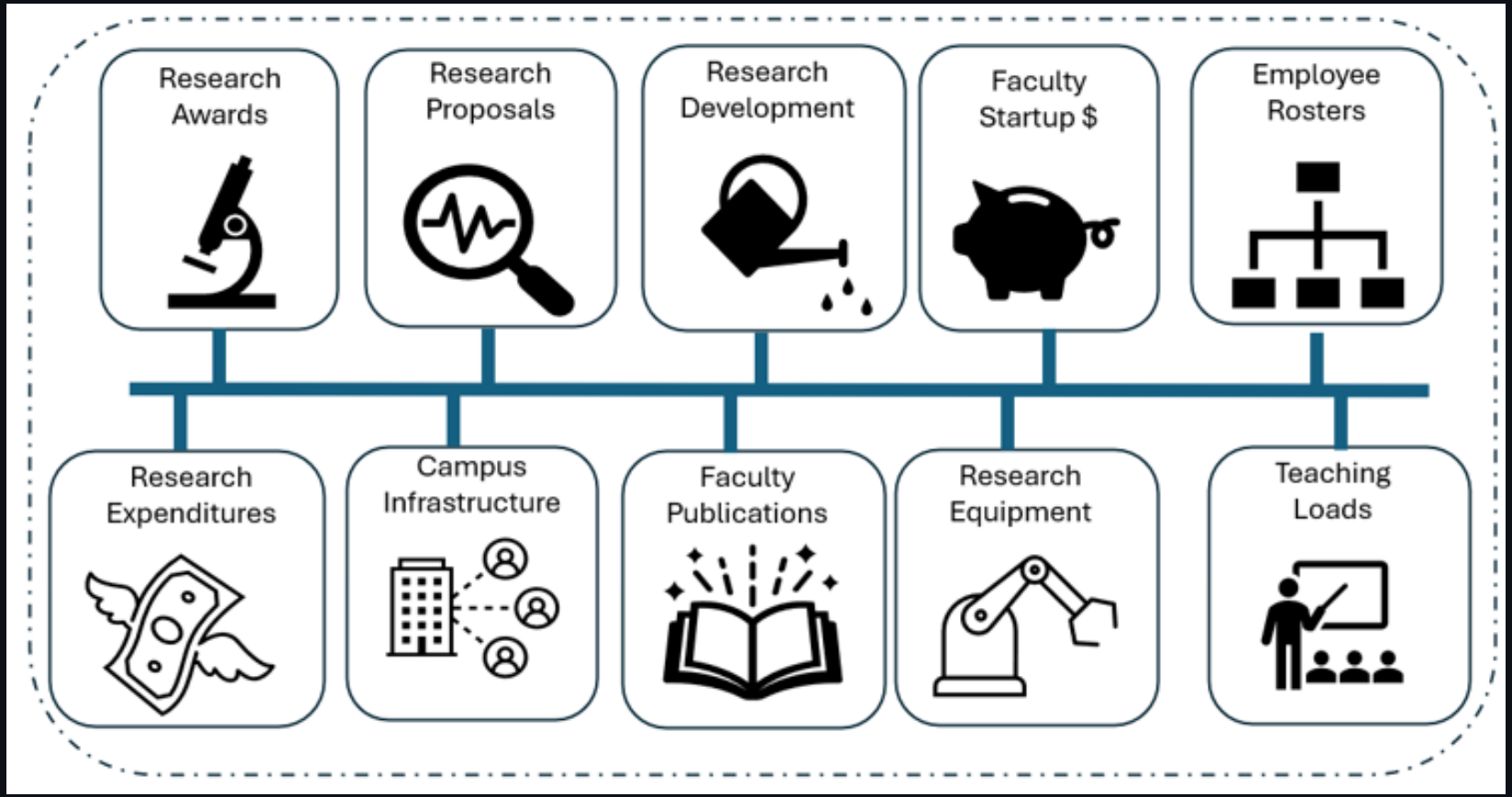






The Data Model

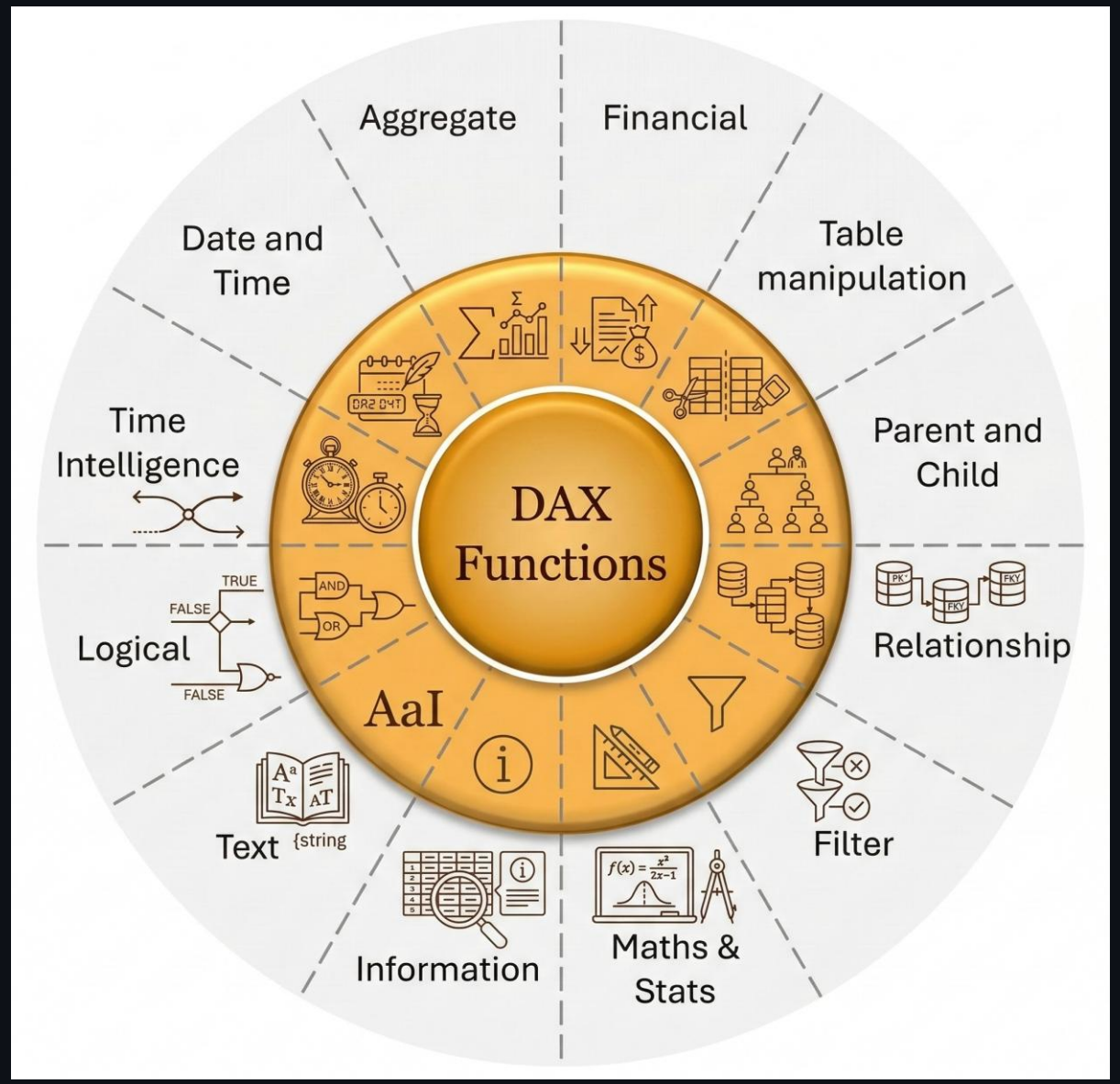
- Relationships NOT merging
- Similar to SQL relationships
- Handles very large data
- Powered by DAX



Data Analysis Expressions (DAX)

- Data Models = BRAINS

  DAX = LOGIC
- Powerful language
- Not coding, it is an expression language
- Built for cross-data analysis
- Examples later



Core DAX Function Groups

- DAX function reference
- We write DAX later
- Date and Time
- Filter
- Time Intelligence

POWER BI

DAX

Deployment Models

- Single report does everything
- NOT scalable, breaks easily
- SPLIT into 2 buckets
- Data Models | Reporting

Is it Time to Split the Data model from the Report?

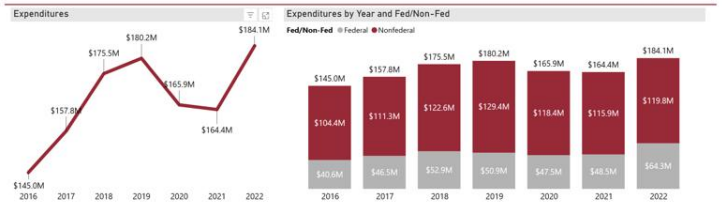
- Multiple reports, same data?
- Easy to refresh?
- What if data source changes?
- Local files or cloud?

Data Model

- Design
- Create

Reporting

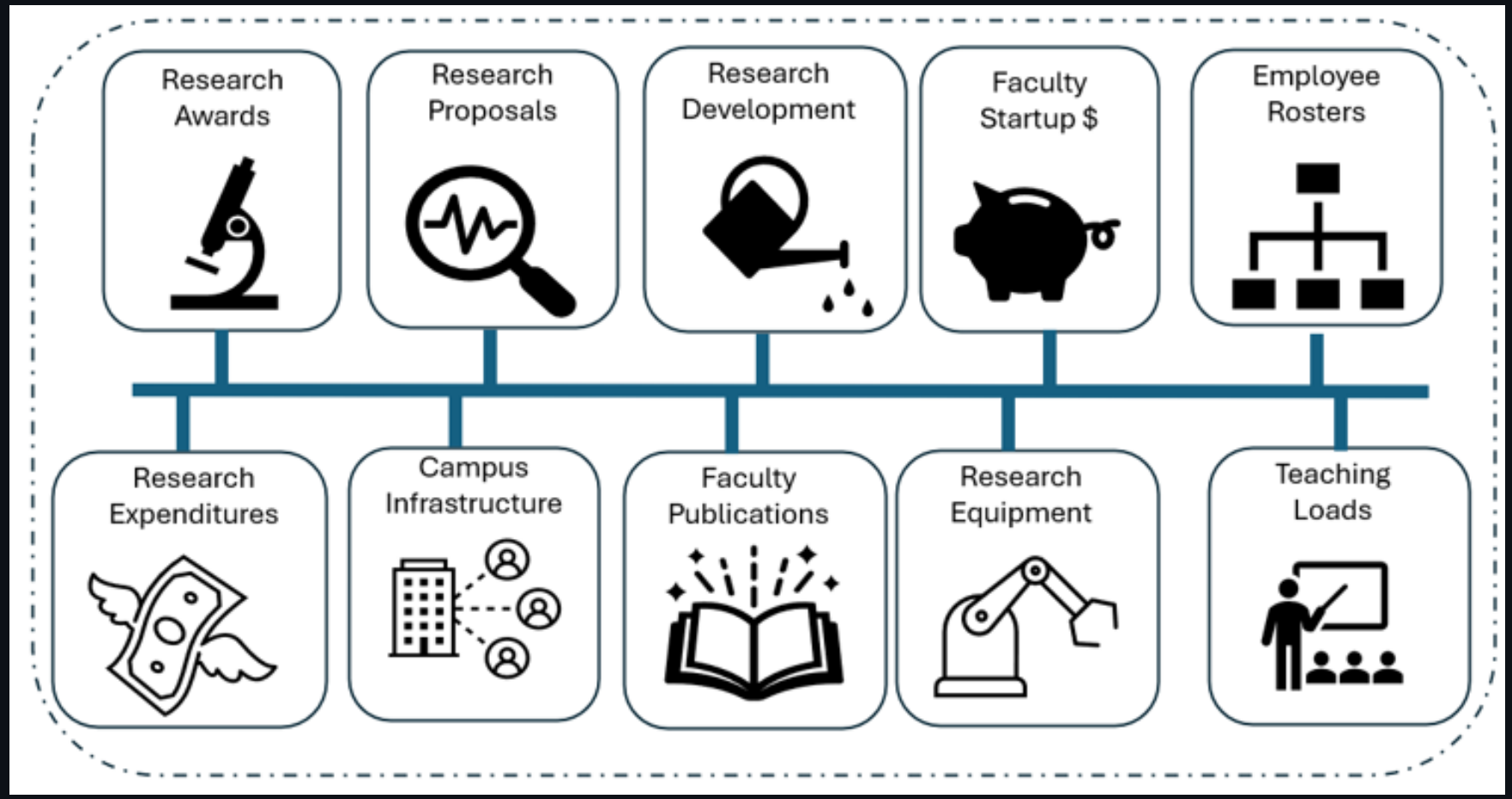
- Create
- Distribute



Fed/Non-Fed	2016	2017	2018	2019	2020	2021	2022	Total
Federal	$40,642,000	$46,490,000	$52,937,000	$50,875,000	$47,454,000	$48,534,000	$64,252,000	**$351,184,000**
Nonfederal	$104,378,000	$111,301,000	$122,561,000	$129,350,000	$118,433,000	$115,909,000	$119,828,000	**$821,760,000**
Total	**$145,020,000**	**$157,791,000**	**$175,498,000**	**$180,225,000**	**$165,887,000**	**$164,443,000**	**$184,080,000**	**$1,172,944,000**

The Data Model as Foundation

- The Brain (Remember connect not merge)

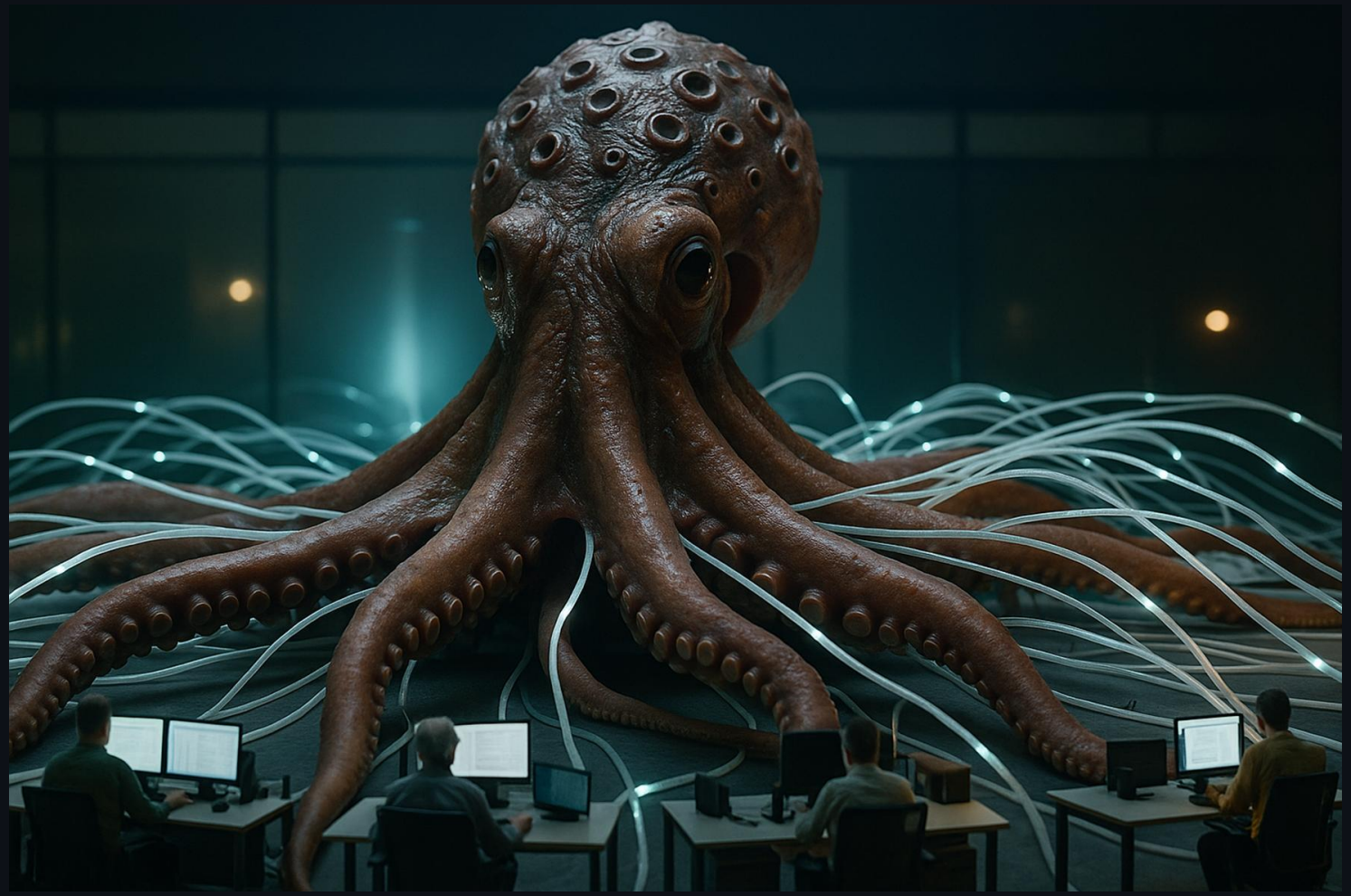


Reporting Layer

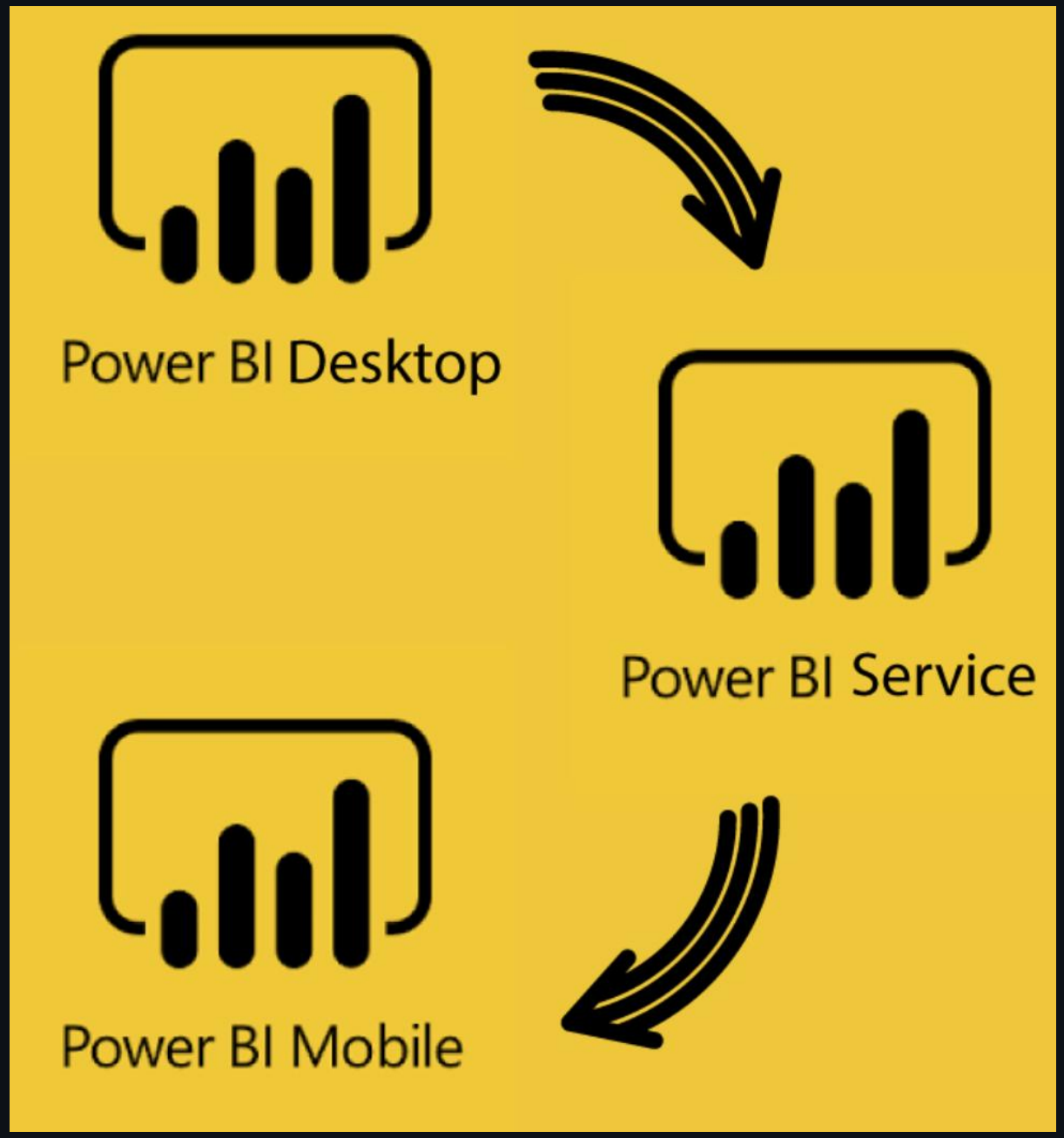
- Multiple reports, one data model
- Same data, different story
  - Executive
  - OSP Director
  - Department Head
  - Grant Specialist

Deploying Reports

- Report >> Users
- Quick one-off share?
- How many users?
- Need to download data?
- Draft or production?



Deployment Options



Power BI Service Workflow

- Desktop >> Workspace >> Service >> APP >> Mobile
- Workspace: Staging area, build and test
- Apps: Polished reports for end-users, access control
- Mobile: Apple and Android

Power BI Workspace	Power BI App
• Used for collaboration • Like a secure, shared folder in the cloud • Specifically designed for BI (Business Intelligence) content	• A curated collection of Reports • Packaged together for distribution • Intended for a broad audience • Created from workspaces • A polished product or Report • Ready for consumption

Let's Build a Report

- Questions?
- We will build on report each session

Live Demo

- Page-1 - Replicate the tables on the left-
- Page-2 - Expandable table - Filter improvements-
- Page-3 - Additional visuals from store

The Data

- Dummy proposals data

Live Demo Page 1

College Totals

Totals

College	Count of PI ID	Count of Proposal Number	Sum of Proposal $
Engineering	226	722	$335,870,924
Arts and Sciences	209	450	$308,339,050
Health Science	51	103	$60,625,223
Education	42	61	$31,246,380
Total	**528**	**1336**	**$736,081,578**

Counts

College	Denied	Funded	Withdrawn	**Total**
Engineering	465	252	7	**722**
Arts and Sciences	301	147	2	**450**
Health Science	79	21	3	**103**
Education	41	20		**61**
Total	**886**	**440**	**12**	**1336**

Counts

College	Denied	Funded	Withdrawn	**Total**
Arts and Sciences	66.89%	32.67%	0.44%	**100.00%**
Education	67.21%	32.79%		**100.00%**
Engineering	64.40%	34.90%	0.97%	**100.00%**
Health Science	76.70%	20.39%	2.91%	**100.00%**
Total	**66.32%**	**32.93%**	**0.90%**	**100.00%**

Live Demo Page 2

Expandable Table | Improve Filter

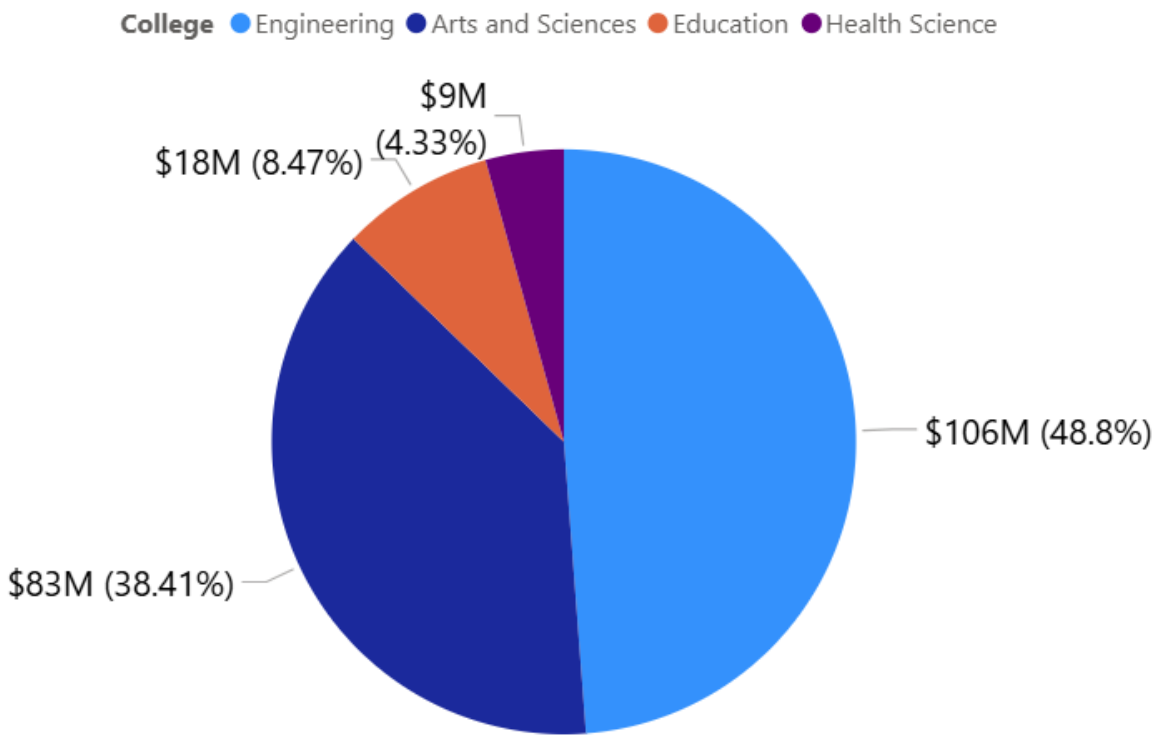
Proposal Status

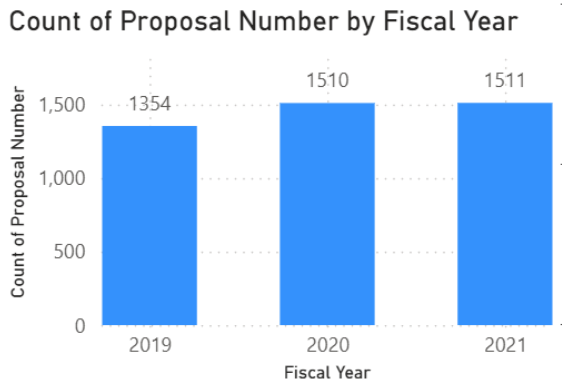
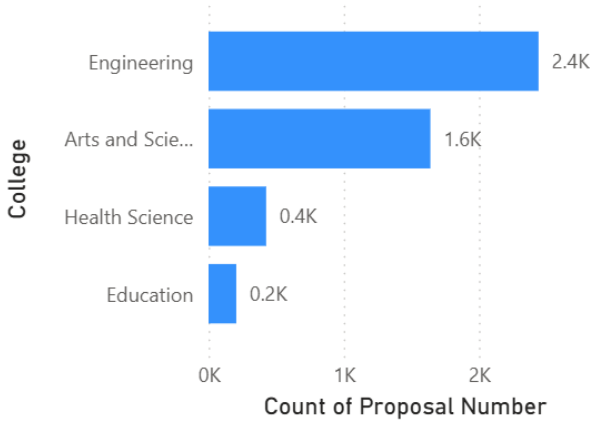
All

Totals

Originating Sponsor Group	Count of PI ID	Count of Proposal Number	Sum of Proposal $
Federal	520	1226	$1,062,260,530
Nonprofit	103	141	$40,812,624
Industry	69	113	$12,409,098
Other	14	14	$2,833,680
Local	7	10	$506,954
State	6	8	$1,005,854
Total	**589**	**1511**	**$1,119,828,739**

Funded Proposal $ by Originating Sponsor Group



Live Demo Page 3

Replicate these charts | Open additional visuals via store